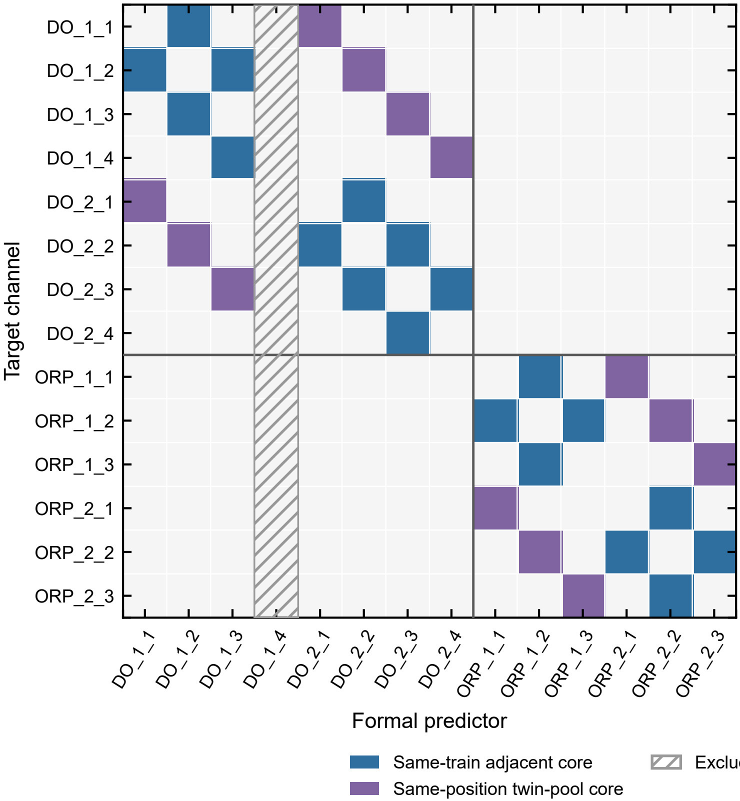
Supplementary Figure S1. Formal PLS peer topology used in D1 drift scoring

(a) Production-active topology matrix



(b) Effective model specification

Target	Formal predictors	p / k
DO_1_1	DO_1_2, DO_2_1	2 / 2
DO_1_2	DO_1_1, DO_1_3, DO_2_2	3 / 3
DO_1_3	DO_1_2, DO_2_3	2 / 2
DO_1_4	DO_1_3, DO_2_4	2 / 2
DO_2_1	DO_1_1, DO_2_2	2 / 2
DO_2_2	DO_1_2, DO_2_1, DO_2_3	3 / 3
DO_2_3	DO_1_3, DO_2_2, DO_2_4	3 / 3
DO_2_4	DO_2_3	**1 / 1**
ORP_1_1	ORP_1_2, ORP_2_1	2 / 2
ORP_1_2	ORP_1_1, ORP_1_3, ORP_2_2	3 / 3
ORP_1_3	ORP_1_2, ORP_2_3	2 / 2
ORP_2_1	ORP_1_1, ORP_2_2	2 / 2
ORP_2_2	ORP_1_2, ORP_2_1, ORP_2_3	3 / 3
ORP_2_3	ORP_1_3, ORP_2_2	2 / 2

Rows are targets and columns are predictors; only production-active links are shown. DO_1_4 is excluded as a predictor, and DO_2_2 to DO_2_4 was rejected in Fig. 11.
Validation depth: DO_2_4 uses the full forward/hold-out/injection audit; other targets use topology-constrained three-fold blocked CV. p = peer count; k = retained components.